

Name: _____

Date: _____

Radical Equations: Challenge Problems

Show all work. Justify your reasoning completely.

Problem 1: Error Analysis (6 points)

A student was asked to solve the equation $\sqrt{x+5} + \sqrt{x} = 5$ and produced the following work:

Jordan's Solution:

$$\sqrt{x+5} + \sqrt{x} = 5$$

$$\sqrt{x+5} = 5 - \sqrt{x}$$

$$(\sqrt{x+5})^2 = (5 - \sqrt{x})^2$$

$$x + 5 = 25 - x$$

$$2x = 20$$

$$x = 10$$

$$\text{Check: } \sqrt{10+5} + \sqrt{10} = \sqrt{25} + \sqrt{10} = 5 + 3.16 \approx 8.16 \neq 5$$

“There is no solution.”

Part A: Identify the specific error in Jordan's work. Explain what Jordan did incorrectly.

Part B: Solve the equation correctly. Show all steps and check your answer.

Part C: Explain why Jordan's error is a common mistake when solving radical equations.

Problem 2: Construct an Equation (6 points)

Part A: Write a radical equation of the form $\sqrt{ax + b} = x + c$ (where a , b , and c are integers) such that:

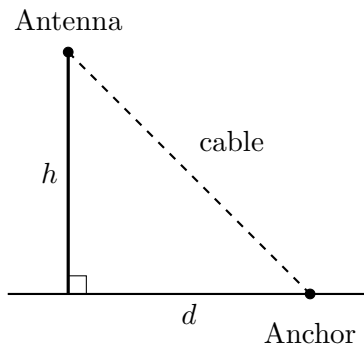
- $x = 5$ is a **valid solution**, AND
- $x = -1$ is an **extraneous solution**

Show the algebraic work that verifies both conditions are met.

Part B: Is it possible to write a radical equation of the form $\sqrt{ax + b} = cx + d$ that has **two valid solutions**? If yes, provide an example and verify. If no, explain why not.

Problem 3: Geometric Application (6 points)

A support cable is attached from the top of a vertical antenna to an anchor point on the ground. The situation can be modeled as shown:



The cable length can be expressed as $\sqrt{h^2 + d^2}$, where h is the height of the antenna in feet and d is the horizontal distance from the base to the anchor point.

Given information:

- The cable is 8 feet longer than the horizontal distance d .
- The height of the antenna is 4 feet less than the horizontal distance d .

Part A: Write an equation in terms of d that models this situation.

Part B: Solve the equation algebraically to find the horizontal distance d .

Part C: Find the height of the antenna and the length of the cable. Verify that your answer makes sense in the context of the problem.

Problem 4: Physics Application (6 points)

In physics, the time T (in seconds) for a pendulum to complete one full swing is given by:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

where L is the length of the pendulum (in meters) and g is the acceleration due to gravity.

On Earth, $g \approx 9.8 \text{ m/s}^2$. On the Moon, $g \approx 1.6 \text{ m/s}^2$.

Part A: Solve the formula for L in terms of T and g .

Part B: A grandfather clock uses a pendulum with a period of exactly 2 seconds (called a “seconds pendulum” because each half-swing takes 1 second). Find the exact length of this pendulum on Earth. Then express your answer as a decimal rounded to the nearest centimeter.

Part C: An astronaut brings a pendulum clock to the Moon. If the pendulum length stays the same as on Earth (from Part B), will the clock run faster or slower on the Moon? Calculate the new period and determine how many seconds the clock would gain or lose per minute.

Problem 5: Parameter Analysis (6 points) – BONUS

Consider the equation $\sqrt{x - k} = x - 4$, where k is a constant.

Part A: Solve the equation for x in terms of k . (You should get two expressions involving k .)

Part B: For the solutions to be *real numbers*, what restriction must be placed on k ? (Hint: Consider what's under the square root in your solutions from Part A.)

Part C: Complete the analysis:

- For what value(s) of k does the equation have **exactly one valid solution**?
- For what value(s) of k does the equation have **two valid solutions**?
- For what value(s) of k does the equation have **no valid solutions**?

Part D: Verify your analysis by testing one specific value of k from each category above.